

Show that the graph G is 2 colorable

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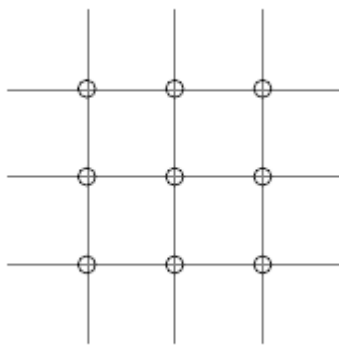


1



Show that the graph G is 2 colorable which is define with vertex set $\mathbb{Z}^2$ where (m, n) is adjacent to (j, k) if $(m, n) \pm (1, 0) = (j, k)$ or $(m, n) \pm (0, 1) = (j, k)$.

I just draw a portion of the graph and it seems easily two colorable, like if I color a vertex with red and a different color (like blue) with all of it's adjacent vertices. But couldn't write it in a formal way.



I was doubt to write it in plain english because there are infintely many vertices to color, it would be nice if anyone show me the right way.

graph-theory

coloring

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asked 1 hour ago



WhyGraph_

59 5

1 Answer

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2



HINT: Consider whether $m + n$ is even or odd.

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answered 1 hour ago



Brian M. Scott

564k 47 652 1084



I see, if I just pick one color for even and another color for odd then the adjacent points in $\mathbb{Z}^2$ are distinguishable.
Thanks @Brian M. Scott – WhyGraph_ 1 hour ago



@WhyGraph_ : That's right. You're welcome. – Brian M. Scott 1 hour ago



@WhyGraph_. That's right. You're welcome. — Brian W. Scott 1 hour ago

